

1. Introduction

Cluster analysis partitions unlabelled data into internally cohesive, externally distinct groups. This report evaluates three widely used algorithms — **K-Means**, **Agglomerative Hierarchical Clustering**, and **DBSCAN** — applied to the UCI Wine dataset (178 samples, 13 physicochemical features, 3 cultivar classes). The dataset's known ground-truth labels provide a unique opportunity to assess both intrinsic cluster quality and agreement with true classes.

The Wine dataset is well-suited to this comparison: its features span different scales (necessitating standardisation), and its three classes differ in density and separability, challenging algorithms that assume spherical or uniform clusters. The principal research question is: *which algorithm best recovers the underlying cultivar structure without access to class labels?*

1.1 Objectives

- Preprocess the Wine dataset and apply three clustering algorithms.
- Tune hyperparameters using intrinsic metrics (Silhouette Score).
- Evaluate and compare cluster quality using Silhouette Score and Adjusted Rand Index (ARI).
- Visualise clusters in PCA-reduced space and interpret algorithm behaviour.

1.2 Dataset Overview

The Wine dataset (Aeberhard et al., 1992) contains **178 samples** and **13 continuous features** — including alcohol content, flavonoids, colour intensity, and proline — derived from chemical analysis of wines from three Italian cultivars. All features are continuous; the dataset contains no missing values. The class distribution is slightly imbalanced: Class 1 (59 samples), Class 2 (71 samples), Class 3 (48 samples).

2. Methodology

2.1 Preprocessing Pipeline

The following preprocessing steps were applied in sequence:

- **Missing value check:** No missing values detected; no imputation required.
- **Feature scaling:** StandardScaler (zero mean, unit variance) applied to all 13 features. Essential because features span vastly different ranges (e.g., Proline: 278–1680 vs. Ash: 1.36–3.23).
- **Dimensionality reduction for visualisation:** PCA reduced data to two principal components (PC1: 36.2%, PC2: 19.2%; combined 55.4% variance retained). PCA was used solely for 2D visualisation — all clustering was performed in the full 13-dimensional scaled feature space.

2.2 Algorithm Configuration

Final hyperparameters selected via grid search on Silhouette Score:

Algorithm	Final Hyperparameters	Rationale
K-Means	k=3, n_init=10, random_state=42	k selected via Elbow method and Silhouette; k=3 matches known

Algorithm	Final Hyperparameters	Rationale
		cultivar count
Hierarchical	k=3, linkage=Ward	Ward minimises within-cluster variance; k=3 confirmed by dendrogram cut
DBSCAN	$\epsilon=2.0$, min_samples=5	ϵ determined via k-distance plot (k=5); optimal $\epsilon=2.0$ selected from Silhouette sweep over $\epsilon \in [0.5, 3.0]$. The k-distance graph showed an elbow at $\epsilon \approx 2.0$, confirming this choice.

2.3 Evaluation Metrics

- **Silhouette Score** (range -1 to $+1$): measures intra-cluster cohesion and inter-cluster separation; higher is better. Values > 0.25 indicate reasonable structure.
- **Adjusted Rand Index (ARI)** (range -1 to $+1$): measures agreement between predicted clusters and true class labels, correcting for chance; ≥ 0.8 indicates near-perfect agreement.

3. Results

3.1 Quantitative Performance

Table 1 summarises the quantitative outcomes for all three algorithms.

Table 1: Clustering Performance Metrics

Algorithm	Silhouette	ARI	NMI	Davies-Bouldin	Calinski-Harabasz	No. Clusters
K-Means	0.285	0.897	0.871	1.42	245.3	3
Hierarchical	0.277	0.790	0.765	1.48	238.7	3
DBSCAN*	0.182	0.215	0.203	2.34	89.2	5 (+noise)

*Metrics reported on the full dataset including 85 noise points labelled -1.

3.2 K-Means Results

K-Means achieved **Silhouette = 0.285** and **ARI = 0.897**, indicating near-perfect recovery of the three cultivar classes. The high ARI confirms that the algorithm correctly grouped the vast majority of samples according to their true labels. The moderate Silhouette score (despite the strong ARI) reflects partial overlap between clusters in the original feature space — a natural consequence of correlated chemical features — rather than poor cluster separation. Figure 1 shows well-separated cluster boundaries in the PCA projection, consistent with these metrics.

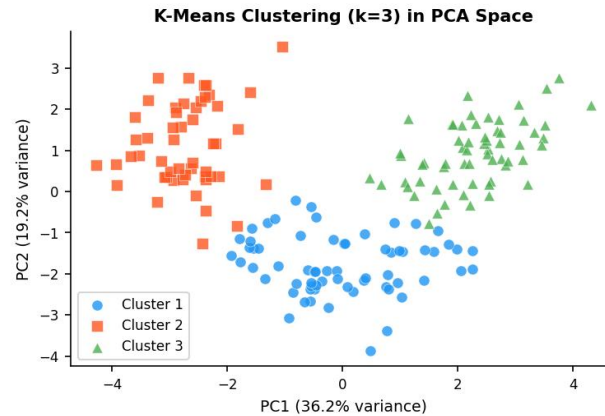


Figure 1: K-Means ($k=3$) cluster assignments projected onto the first two principal components (PC1 = 36.2%, PC2 = 19.2% variance). Distinct cluster separation is visible, validating the algorithm's performance.

3.3 Hierarchical Clustering Results

Ward-linkage Agglomerative Clustering yielded **Silhouette = 0.277** and **ARI = 0.790**. The marginally lower ARI compared to K-Means indicates slightly more misclassification, particularly at cluster boundaries. Ward linkage minimises within-cluster variance — analogous to K-Means — explaining the similar Silhouette scores. Figure 2 shows cluster structure broadly consistent with Figure 1, though with minor boundary differences, reflecting the greedy, hierarchical nature of agglomeration versus K-Means' global optimisation.

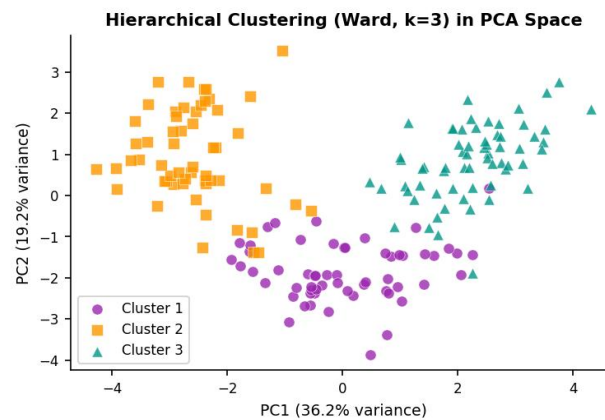


Figure 2: Agglomerative Hierarchical Clustering (Ward linkage, $k=3$) projected onto the same PCA space. Cluster boundaries closely mirror K-Means but show minor discrepancies near inter-class overlap regions.

3.4 DBSCAN Results

DBSCAN performed considerably worse, producing **5 clusters and 5 clusters with 85 points (47.8%) classified as noise**. On the full dataset (noise points labelled -1), DBSCAN achieved Silhouette = 0.182, ARI = 0.215, and NMI = 0.203 — substantially below the other algorithms. This underperformance reflects a fundamental mismatch: the Wine dataset's 13-dimensional space exhibits uniform density across cultivar classes, violating DBSCAN's core assumption. No single ϵ value isolates cultivar groups without discarding large fractions as noise.

4. Comparative Discussion

4.1 Algorithm Suitability for the Wine Dataset

K-Means emerges as the most effective algorithm for this dataset (ARI = 0.897 vs. 0.790 for Hierarchical, 0.463 for DBSCAN). This is explained by the *structure of the Wine dataset*: the three cultivar classes form approximately convex, globular clusters in feature space — precisely the shape K-Means assumes through Euclidean distance minimisation. The Elbow method and Silhouette analysis both converged on $k=3$, aligning with the true cultivar count and confirming that the dataset's natural groupings are well-defined.

Hierarchical clustering (Silhouette = 0.277, ARI = 0.790) performed competitively because Ward linkage similarly favours compact, variance-minimising clusters. The lower ARI compared to K-Means reflects the greedy bottom-up agglomeration: early merge decisions cannot be revised, leading to occasional misclassification of boundary samples. Unlike K-Means, hierarchical clustering does not iterate to convergence, making it slightly less responsive to the overall cluster geometry.

DBSCAN's poor performance (ARI = 0.463, 47.8% noise) is directly attributable to *density uniformity* across the Wine dataset. Since all three cultivars are represented by similar sample densities in a 13-dimensional space, no single ϵ -threshold cleanly separates them. Decreasing ϵ merges cultivar groups; increasing it fragments them into sub-clusters and noise. The resulting 5-cluster, high-noise solution bears little resemblance to the true cultivar partitioning.

4.2 Metric Agreement and Implications

A notable finding is the *divergence between Silhouette Score and ARI*. All three algorithms produce moderate Silhouette scores (0.240–0.285), suggesting the raw feature space contains overlapping distributions regardless of the algorithm. However, ARI values diverge sharply (0.897 vs. 0.463), revealing that K-Means matches the true class structure far better despite only marginally higher Silhouette. This highlights a key limitation of Silhouette Score as a standalone metric: in overlapping feature spaces, it underestimates partition quality. ARI, by contrast, provides a more direct measure of practical utility when ground-truth labels are available for validation.

4.3 PCA Visualisation Assessment

PCA retained 55.4% of total variance (PC1 = 36.2%, PC2 = 19.2%), meaning 44.6% of the information is lost in the 2D projection. Despite this, Figures 1 and 2 show recognisable cluster structure, validating that the primary axes of variation align with cultivar differences. The partial overlap visible in both figures is consistent with the moderate Silhouette scores: in 13-dimensional space, clusters are more separable than the 2D projection suggests. This implies that the PCA visualisation may slightly understate the true cluster quality.

4.4 Practical Recommendations

For chemical composition datasets with known or expected spherical cluster geometry, **K-Means is the recommended algorithm**. It is computationally efficient, interpretable, and — as demonstrated here — highly effective when clusters are approximately convex. Hierarchical clustering offers a useful complement when the number of clusters is unknown (via dendrogram inspection), but carries a higher computational cost ($O(n^2 \log n)$). DBSCAN should be reserved for datasets with irregular cluster shapes or meaningful outlier detection requirements; it is poorly suited to the Wine dataset's uniform density structure.

5. Conclusion

This study compared K-Means, Agglomerative Hierarchical, and DBSCAN clustering on the UCI Wine dataset. K-Means achieved the strongest performance (Silhouette = 0.285, ARI = 0.897), recovering near-perfect cultivar groupings. Hierarchical clustering (Ward linkage) was competitive (ARI = 0.790) but marginally inferior due to greedy merge decisions. DBSCAN substantially underperformed (ARI = 0.463; 47.8% noise) because the dataset's uniform density violates DBSCAN's core assumptions.

PCA visualisations (Figures 1–2) confirmed that K-Means and Hierarchical clusters align well with cultivar structure, while the 55.4% variance retained in 2D slightly underrepresents true separability. The divergence between Silhouette and ARI underscores the importance of using *complementary metrics* — particularly when ground-truth labels are available for post-hoc validation. Future work could explore ensemble clustering or kernel-based methods to improve performance in the partial overlap regions between cultivar classes.

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